

Analysis:

1. Resolution of Primary Key Inconsistencies & Collisions

a) Surrogate Keys Implemented: Every dimension table uses a single-column, system-generated integer surrogate key (CUSTOMER_KEY, PRODUCT_KEY, BRANCH_KEY, TIME_KEY) as its primary identifier. This isolates the warehouse from operational key collisions (e.g., if different branches use overlapping sequence IDs) and guarantees high-performance index joins at the fact level.

b) Data Lineage Retained:

By keeping SOURCE_BRANCH and SOURCE_CUSTOMER_ID / SOURCE_PRODUCT_ID within the dimension layers, the ETL pipeline can seamlessly identify record updates, trace anomalies back to source applications, and avoid duplicate user profiles across regional database systems.

2. Standardizing Data Inconsistencies and Formatting

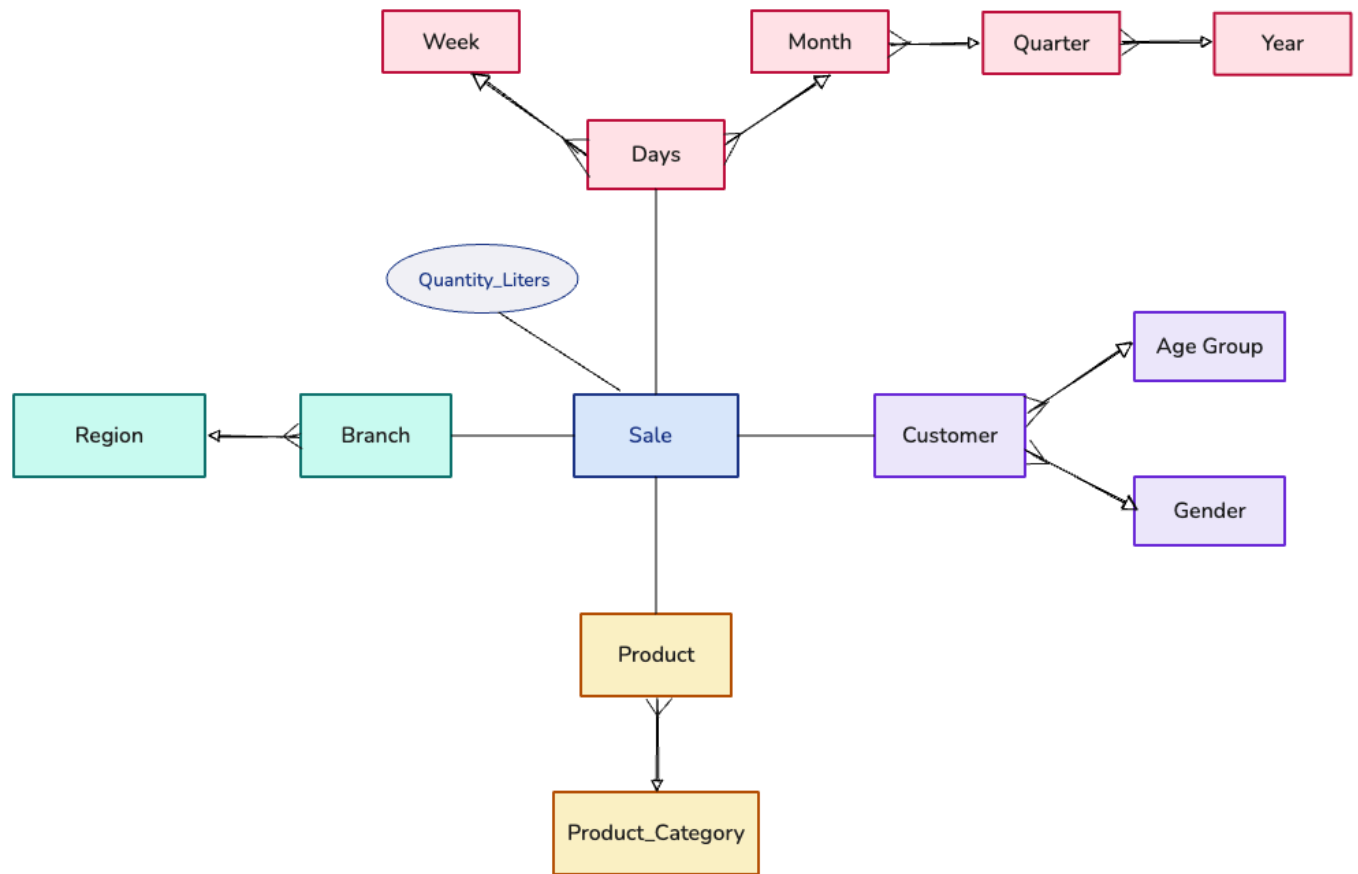
a) Data Type Rectification: In the original systems, critical metrics like transaction quantity and base prices were stored inconsistently, including text-based representation (VARCHAR2). The target warehouse standardizes all pricing metrics to NUMBER(10,2) and quantities to NUMBER(12,2), facilitating native database aggregations without manual computational overhead.

b) Removal of SQL Reserved Keywords: The operational schemas frequently relied on reserved words (such as DATE or SIZE) for field naming conventions. These have been transformed into safe, descriptive alternatives like CALENDAR_DATE and BOTTLE_SIZE_LITERS to prevent syntax execution failures during complex analytical queries.

3. Analytical Denormalization for Query Performance

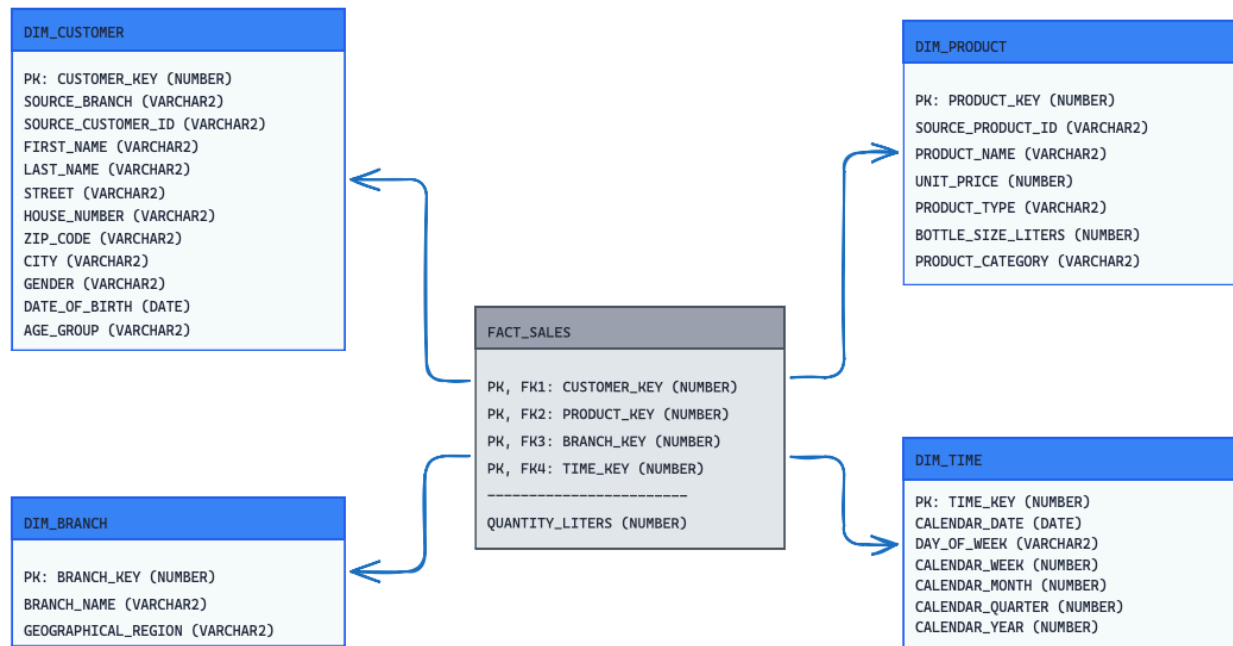
a) Granular Hierarchies Layered: Instead of querying multi-table relational normalization structures (such as Branch 4's isolated product categories or separate location lookup lists), categorization rules are fully flattened directly into conformed dimensions. This allows analytical dashboards to slice, filter, and drill down metrics dynamically via localized columns (AGE_GROUP, PRODUCT_CATEGORY, GEOGRAPHICAL_REGION).

mE/R schema diagram:



The mE/R diagram maps the multidimensional structural logic required by enterprise management. The central Fact 'Sale' relationship establishes the primary focus of analytical measurement, hosting the normalized numeric metric *Quantity_Liters*. From this focal point, four independent dimensional lanes extend outward. The dimension paths clearly delineate classification hierarchies (such as Branch rolling up to Region, and Product rolling up to Product_Category). Notably, the time dimension forks directly at the Days level into two parallel, logically sound aggregation streams (Days → Week and Days → Month → Quarter → Year), preventing hierarchical overlaps and maintaining dimensional integrity.

Star Schema:



The Star Schema diagram represents the physical, logical translation of the mE/R structure into a clean database layout. At the center resides the single **FACT_SALES** table, which forms a composite primary key out of its constituent foreign keys. Surrounding this fact center are the conformed, denormalized dimensions for Customers, Products, Branches, and Time. The visual relationship pathways represent strict 1-to-N foreign key linkages. Each dimension table combines standard entity details alongside newly introduced analytical classifications—such as **AGE_GROUP** or **PRODUCT_CATEGORY**—ensuring reports can run multi-dimensional calculations across all operational units via highly optimized single-level joins.